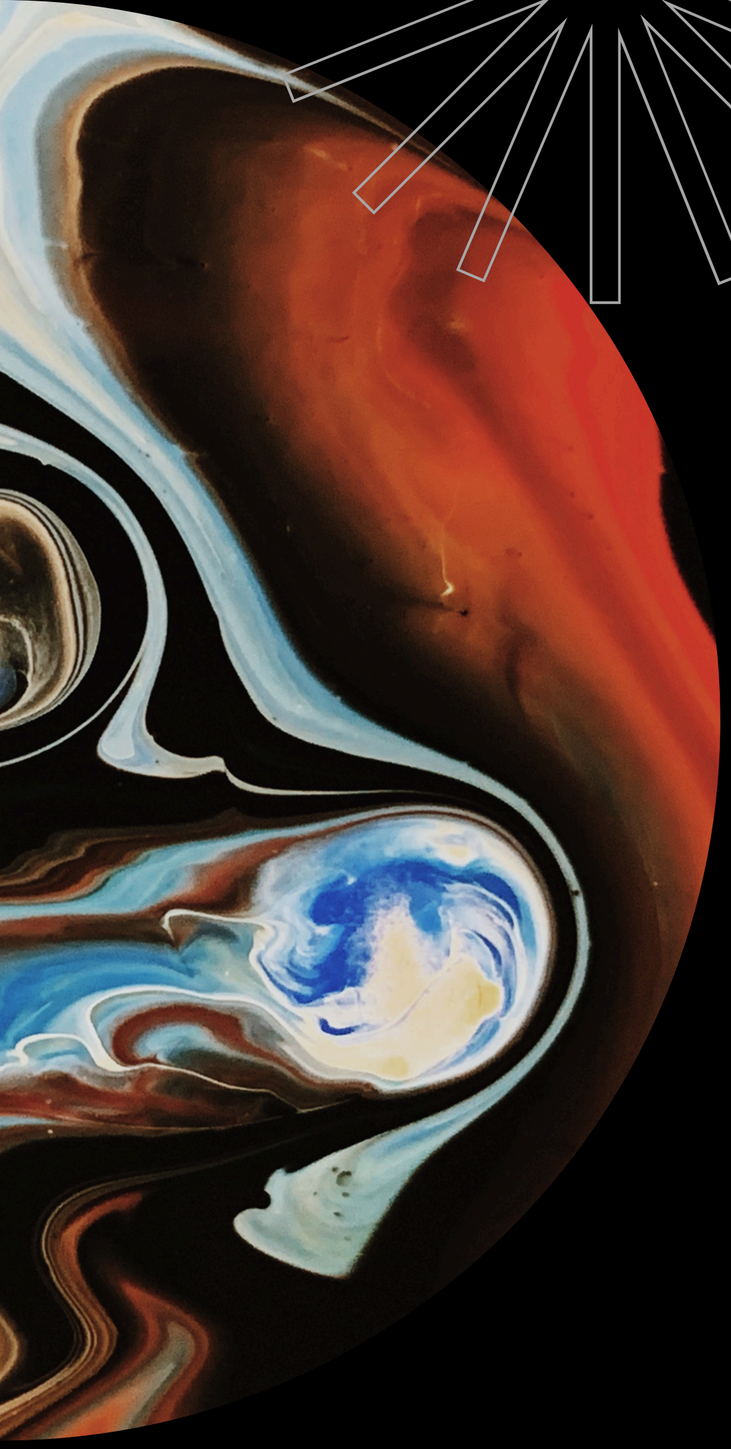
PORTFOLIO

MACHINE LEARNING · DATA · SOFTWARE





TANUSH THAKRAN

Computer Science undergraduate at BITS Pilani working across applied machine learning, data analysis and software engineering. Built a learning-to-rank model on 3.35M rows with leakage-safe evaluation, a fully offline RAG pipeline that cites the video and timestamp of every answer, and a Power BI analysis of the public Superstore dataset that traced a margin problem to a specific discount threshold. Comfortable owning a problem end to end, from raw data to recommendation or from schema to deployment.

INTRODUCTION



A behaviour-cloning model that predicts which legal move a winning agent chooses in 4,604 recorded Pokémon TCG AI battles. Each decision is a variable-length menu of options, so it is framed as learning-to-rank rather than classification.

Pokémon TCG Move Prediction

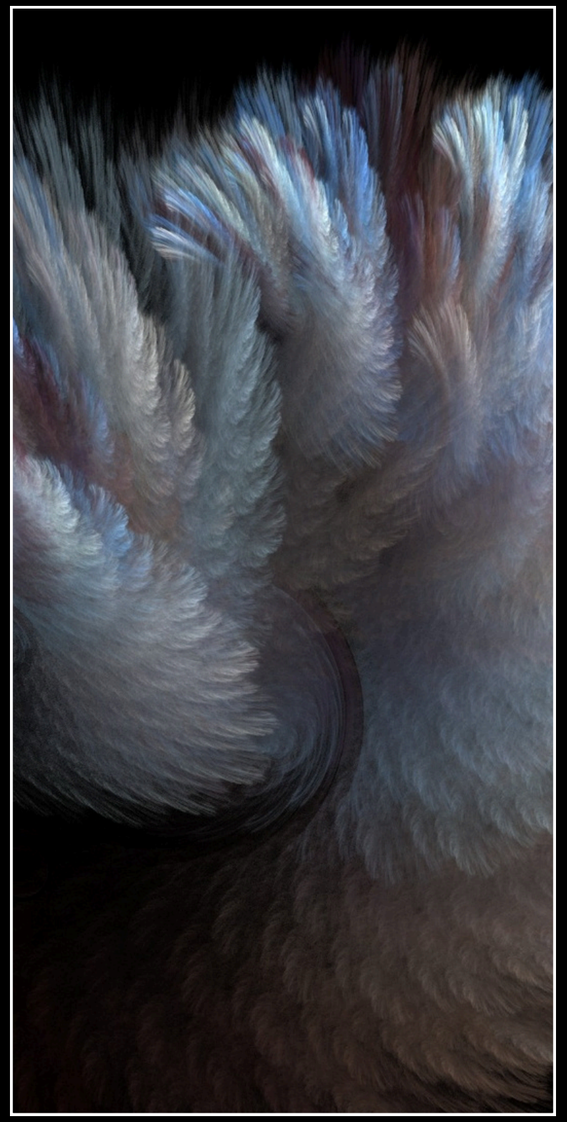
GitHub Repo Link :
github.com/Tanush1206/pokemon-tcg-move-prediction

KEY FEATURES

- **Learning-to-Rank, Not Classification:** LightGBM LGBMRanker with a LambdaRank objective, trained on 3.35M rows and 83 features.
- **Leakage-Safe Splits:** 70/15/15 split by episode, never by row, so turns from one game never appear on both sides.
- **Measured Result:** 57.4% top-1 accuracy on 50,649 held-out decisions vs. a 21.7% random baseline (2.65x). The true move is in the top 3 88% of the time.
- **Ablation-Driven:** A 7-run experiment log on a fixed test set. The largest gain (+0.054) came from reverse-engineering undocumented option indices to resolve which card each option refers to.
- **Negative Result Reported:** A matched-volume control showed that filtering for "stronger" episodes did not help.
- **Fast Extraction:** 4,604 game files streamed into one Parquet file in about 75 seconds on 31 workers.

KEY FEATURES

- **End-to-End Ingestion:** ffmpeg audio extraction, then Whisper large-v2 transcription with translation into timestamped segments.
- **Semantic Retrieval:** Each segment is embedded with local bge-m3. The question is embedded the same way and the top 5 segments are retrieved by cosine similarity.
- **Every Answer Traceable:** Each Whisper segment is one chunk, so every retrieved passage maps to an exact moment in a numbered, titled video.
- **Grounded Generation:** The prompt makes deepseek-r1 (via Ollama) answer only from retrieved segments, cite video, title and timestamp, and say so when the answer isn't there. This is designed to reduce hallucination and to keep answers traceable.
- **Scale Debugging:** Embedding a long course crashed the Ollama runner because single requests were too large. Sending segments in fixed-size batches fixed it.
- **Fully Local:** No data leaves the environment, and there is no API cost.



RAG BASED AI

A fully offline retrieval-augmented QA pipeline that turns Hindi-language course videos into a searchable English knowledge base. Every model runs locally, with no external APIs. Answers are grounded in retrieved transcript segments and cited to video, title and timestamp.

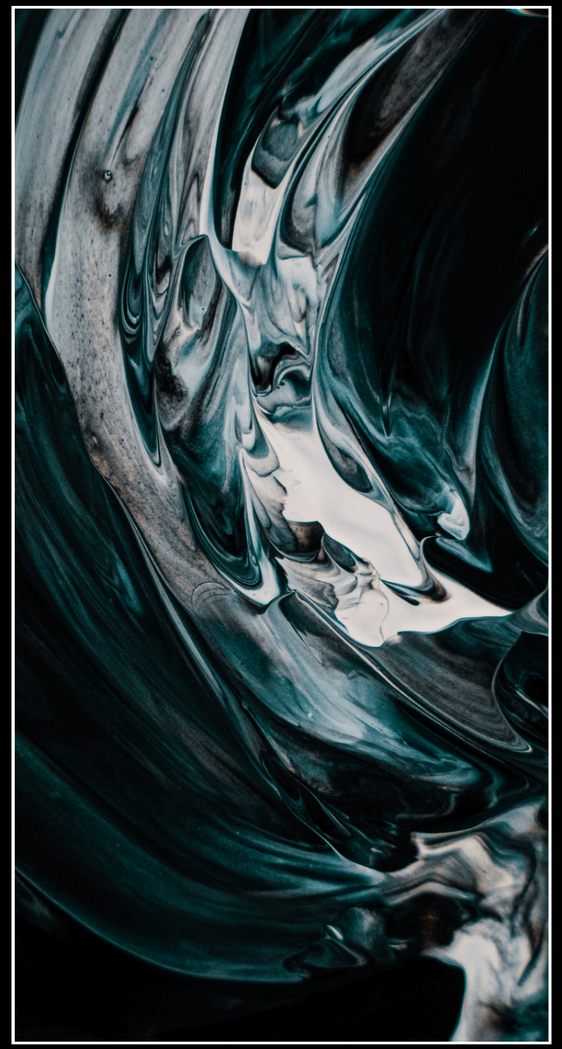
GitHub Repo Link :
github.com/Tanush1206/rag_based_ai

LLM UTILITY LAB

A modular LLM application for text summarization and context-aware Q&A, built in Python on the Groq API through its OpenAI-compatible interface. It is served as a FastAPI service in Docker and deployed on Render.

Github Repo Link :
github.com/Tanush1206/llm-utility-lab

Live Link :
llm-utility-lab.onrender.com/docs



KEY CONTRIBUTIONS

- **Modular Design:** Client communication, prompting, summarization, Q&A, response modelling and evaluation are independent modules.
- **Token Tracking:** Input, output and total tokens are recorded for every request.
- **Grounded Q&A:** Answers are tied to user-provided context to reduce unsupported answers.
- **Evaluation Runner:** Scores live responses against three predefined cases by normalised text match, saves each run with model, temperature and prompt version, and flags any case that regresses from pass to fail between runs.
- **Engineering Hygiene:** 58 pytest tests (the API client is faked, so tests make no live calls) run in GitHub Actions CI. Model errors return 502, unexpected errors 500, and the request timeout is configurable.



An end-to-end telecom churn system on the IBM Telco dataset (7,043 customers), built with scikit-learn and deployed as a Streamlit app.

Customer Churn Prediction

GitHub Repo Link :
github.com/Tanush1206/Customer-Churn-Prediction

Live Link :
customer-churn-prediction.streamlit.app/

KEY FEATURES

- **Leakage-Safe Pipeline:** Scaling and encoding go through a ColumnTransformer inside a pipeline. Logistic Regression, Random Forest and Gradient Boosting were compared on held-out data.
- **Model Choice Justified:** Logistic Regression was kept because it was within 0.003 ROC-AUC of Gradient Boosting and directly interpretable. Regularisation was tuned with 5-fold CV.
- **Threshold From Out-of-Fold Predictions:** The 0.30 threshold maximises F1 on out-of-fold training predictions, instead of the default 0.50.
- **Measured Impact:** On held-out data, recall rose from 55.9% to 75.1% at 51.8% precision, with 84.1% ROC-AUC.
- **Explainable:** The app shows each feature's contribution to a customer's risk score.

KEY FEATURES

- **Root-Cause Analysis, Not Reporting:** Tested whether a margin problem stemmed from product mix or pricing, rather than presenting charts without a conclusion.
- **Validated Across Four Cuts:** The discount-margin relationship was checked at sub-category, region, category and customer level before drawing a conclusion.
- **Quantified Recommendation:** Central region averages a 24.0% discount for a 7.9% margin, against West's 10.9% discount for a 14.9% margin. Closing that gap is worth an estimated $35K over the four years of data, assuming order volume holds, with a single-region test proposed before rollout.
- **Loss-Making Products:** Tables (−$17.7K), Bookcases (−$3.5K) and Supplies (−$1.2K) lose money. Tables has the highest average discount, 26.1%, and the recommendation is to cap Tables and Bookcases at 20%.
- **Peak-Season Dilution:** November is one of the strongest sales months, but its margin falls to 10.06% against 12.47% for the year.
- **Interactive Power BI Dashboard:** DAX measures, cross-filtering slicers, and conditional formatting that surfaces loss-making sub-categories at a glance.
- **Data Integrity First:** Handled a Windows-1252 encoding failure and a silent US-locale date-parsing trap, then reconciled every load against independently computed control totals before analysis.
- **SQL Analysis:** PostgreSQL queries (CASE bucketing, window functions, cohort retention, NTILE-based RFM segmentation) reproduce the dashboard's numbers.



SUPERSTORE SALES & PROFITABILITY ANALYSIS

A Power BI and PostgreSQL project on the public Sample Superstore dataset (9,994 line items, 5,009 orders, $2.3M revenue, 2014 to 2017). Revenue grew 51%, so why didn't profit follow? The analysis traced it to discount policy: margin holds at 29.5% undiscounted, turns negative above 25%, and reaches −77% past 40%.

GitHub Repo Link :
github.com/Tanush1206/superstore-powerbi-analysis

VIDEOCAPTION MAKER

A private, offline captioning app, released and installed with a single command. Speech recognition (faster-whisper) and translation (M2M100) run on the user's own GPU or CPU, so the video never leaves the machine. Nine languages, including Hinglish.

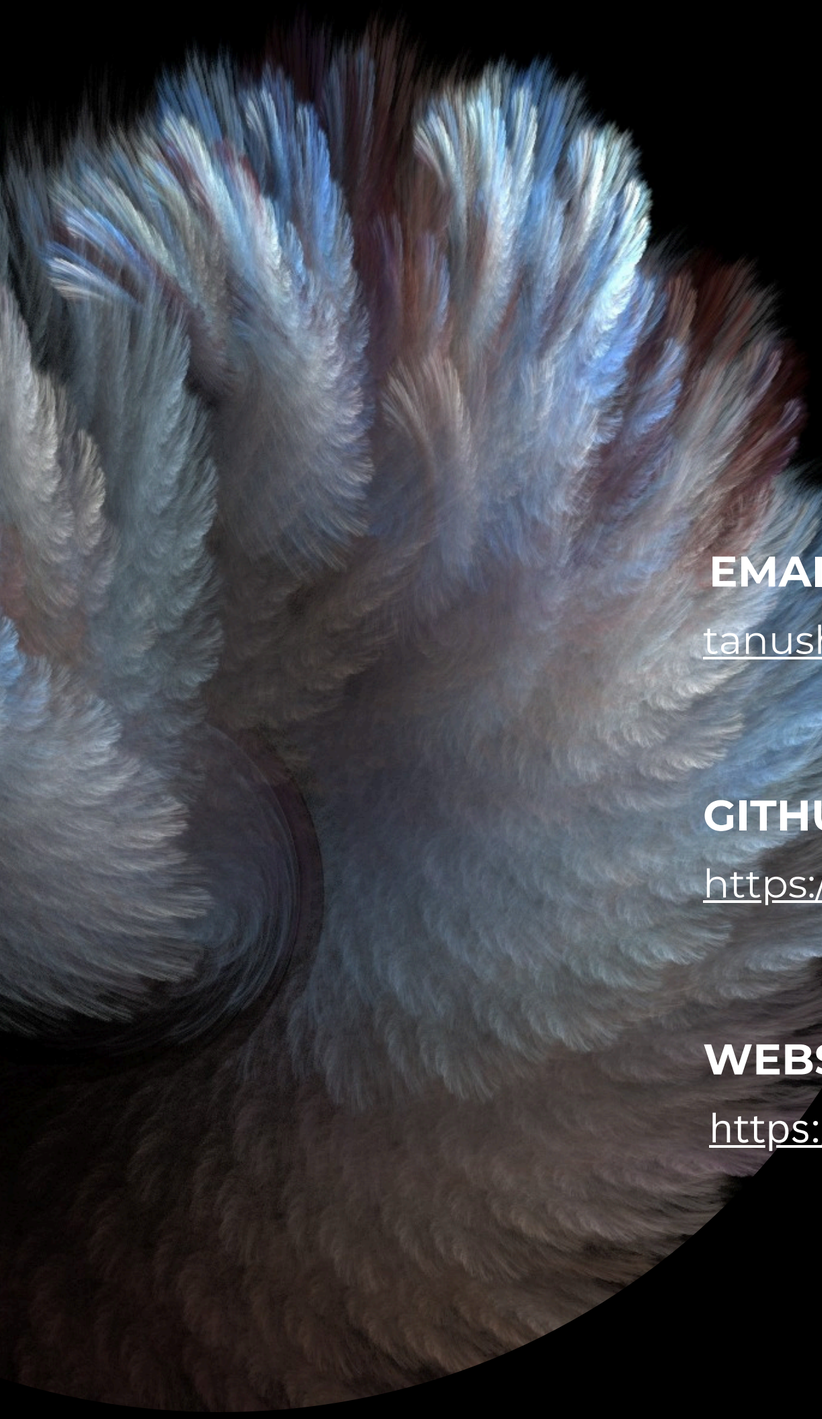
Github Repo Link :
github.com/Tanush1206/Video-Caption-Maker

Live Link :
tanush1206.github.io/Video-Caption-Maker



KEY CONTRIBUTIONS

- **Local Transcription and Translation:** faster-whisper and M2M100 chosen automatically for the hardware (large-v3 on GPU, turbo or small on CPU), with a Celery worker keeping long inference off the API.
- **Caption Editor, Search and Export:** Edit, retime and style captions, search everything said by meaning (ChromaDB), then export burned-in MP4 or SRT/VTT.
- **Measured Language Quality:** Tested on Google's FLEURS speech set: 3-7% word error on European languages, 26% on Hindi. The run exposed a silence-detection bug on quiet recordings (English 59% to 6% after the fix) and led to removing four languages that scored 63-122%.
- **Shipping and Updates:** One-command installer for Windows and Linux (macOS in testing) with SHA-256-verified downloads, GitHub Actions releases to GHCR, and small updates (a typical release is about 6 MB).
- **Reliability:** Interrupted jobs are requeued automatically with a 3-attempt limit, and a request guard blocks DNS-rebinding and cross-origin calls and keeps an audit log.
- **Tested:** 490+ backend tests.
- **Stack:** Next.js, FastAPI, Celery, Redis, PostgreSQL, ChromaDB, Docker Compose.



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